

Class Test

Q1

a Definition:-

Data mining is the process of discovering useful patterns, relationships, trends, and knowledge from large data using statistical, machine learning and database techniques.

Two tools used: $\rightarrow$ classification $\rightarrow$ clustering

KDD (Knowledge Discovered in Database) Process:-

- * Data collection
- Data cleaning
- Data integration
- Data selection
- Data transformation
- Data mining
- Pattern evaluation
- Knowledge presentation

Role of data processing:-

- Remove missing and noisy data
- Improves data accuracy
- Increases mining accuracy
- Reduces processing time.

b Data cleaning:- Data cleaning is the process of detecting and correcting inaccurate, incomplete or inconsistent data.

Common data quality issues

- missing values
- Duplicate records

Handling missing values

- Ignore the record
- Replace with mean/median/mode
- Predict using machine learning.

19
100

Handling Noisy Data:-

- Binning
- Regression
- Clustering

Example:-

marks = 45, 46, 47, 95

95 is an outlier and can be smoothed using binning.

2

a Definition:-

Data integration combines data from multiple sources into a unified dataset.

Challenges

- Data redundancy
- Data inconsistency

Handling Redundancy

- Remove duplicate records
- Use correlation analysis.

Handling inconsistency

- Standardize formats.
- Resolve conflicting values.

Ex:-

Date:

10/02/2025

2025-02-10

convert both into one standard format.

Q what is covariance Analysis? List the two covariance formulas

Covariance Analysis

covariance measures how two variables change together.

Formula - 1 (Population)

$$\text{Cov}(X, Y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

Formula - 2 (Sample)

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Q₃ suppose two stocks A & B have the following values in one week:

A	2	3	5	4	6
B	5	8	10	11	14

observation:

when stock A increases, stock B also increases.

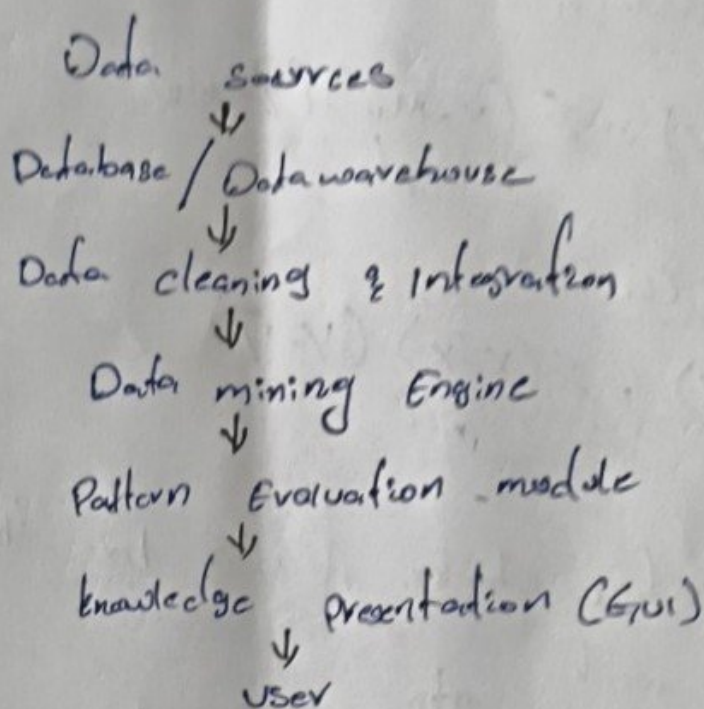
Therefore,

- Positive covariance
- Prices rise together
- they are not independent

The stocks move together due to the same industry trend.

b Explain the architecture of a typical Data mining system with a need diagram.

Diagram



Components

- Data sources: Raw data
- Database / Data warehouse: stores integrated data
- PreProcessing: cleans and transforms data
- Data mining Engine: Applies mining algorithm
- Pattern Evaluation: Finds useful pattern
- Knowledge presentation: Displays results.

Interaction:

Each component processes data sequentially to discover useful knowledge.

Q. Find Covariance

a Given

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

mean of X

$$\bar{X} = \frac{4+5+6+7+9}{5} = 6.2$$

mean of Y

$$\bar{Y} = \frac{8+3+7+5+6}{5} = 5.8$$

X	Y	$X - 6.2$	$Y - 5.8$	product.
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

$$\text{Total product} = -1.8$$

$$\text{Sample covariance} = \frac{-1.8}{4} = -0.45$$

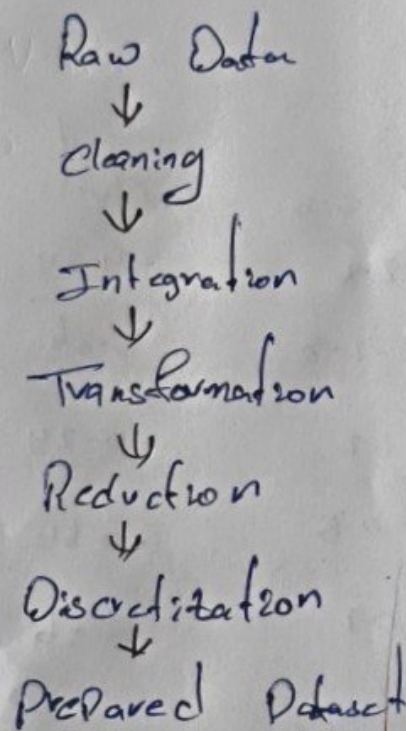
$$\text{Population covariance: } \frac{-1.8}{5} = -0.36$$

b Describe the major steps in Data preprocessing.

major steps

- Data cleaning
- Data integration
- Data Transformation
- Data Reduction
- Data Discretization

Pipe Line:-



Justification :-

cleaning removes errors ~~from~~, integration combines sources, transformation normalizes values, reduction decreases size and discretization converts continuous data into intervals.

Q5 a Equal-Frequency (Equi-Depth) Binning

Data:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Total values = 27

choose 3 equal-frequency bins

Each bin = 9 values

Bin-1

13, 15, 16, 16, 19, 20, 20, 21, 22

$$\text{mean} = \frac{13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22}{9} = \frac{162}{9} = 18$$

~~Bin~~ smoothed Bin = 18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2 :- 22, 25, 25, 25, 25, 30, 33, 33, 35

$$\text{mean} = \frac{22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35}{9} = \frac{253}{9} = 28.1$$

smooth bin :- 28.11 $\rightarrow$ 9-times

Bin-3 :- 35, 35, 35, 36, 40, 45, 46, 52, 70

$$\text{mean} = \frac{35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70}{9} = 43.78$$

smooth bin :- 43.78 $\rightarrow$ 9-times

b Explain Data Discretization. compare Equal-width and Equal-Frequency Binning.

Data Discretization:-

Data discretization converts continuous values into intervals (bins)

Importance:-

- Reduce noise.
- Simplifies analysis
- Improves classification

Comparison

Features	<u>Equal-width</u>	<u>Equal-Frequency</u>
Interval size	Equal	Different
No. of records	Different	Equal
sensitive to outliers	Yes	Less
Best for	uniform data	skewed data

Ex- Data :- 10, 20, 30, 40, 50, 60

Equal width

10 - 20

31 - 50

61 - 70

Equal-Frequency

10, 20

20, 30

30, 40

50, 60

Impact

Equal-width is simple but affected by outliers.

Equal-frequency balances records bins & usually provides better smoothing.